

SCIENCE EDUCATION

# PHYSICS

A Comprehensive Guide for Class XI

*Premium Edition • CBSE Curriculum*

# Motion in a Plane

*Vectors, Projectile Motion, and Circular Motion*

In the previous chapter, we studied motion along a straight line — one-dimensional motion. However, most real-world motions occur in two or three dimensions: a ball thrown at an angle follows a curved path, the Earth orbits the Sun in an elliptical plane, and a car navigating a roundabout moves in a circle. To describe such motions, we need a mathematical tool that can handle both magnitude and direction simultaneously — **vectors**.

This chapter introduces vectors and their algebra, then applies them to two important types of two-dimensional motion: **projectile motion** (the curved path of objects launched near Earth's surface) and **uniform circular motion** (motion along a circular path at constant speed). These concepts are fundamental to everything from sports science to satellite orbits.

## Historical Background: From Arrows to Algebra — The Story of Vectors

The concept of quantities with direction dates back to ancient mechanics, but formal vector mathematics took centuries to develop.

**Simon Stevin (1548–1620)**, a Flemish engineer, was among the first to use the parallelogram law of forces around 1586 — long before the word "vector" existed. He demonstrated that three forces in equilibrium form a closed triangle.

**Sir William Rowan Hamilton (1805–1865)**, an Irish mathematician, coined the term "vector" (from the Latin *vehere*, meaning "to carry") in the 1840s while developing his quaternion algebra. **Josiah Willard Gibbs (1839–1903)** of Yale and **Oliver Heaviside (1850–1925)** of England independently simplified Hamilton's quaternions into the modern vector algebra — with  $\hat{i}$ ,  $\hat{j}$ ,  $\hat{k}$  notation, dot product, and cross product — that physicists use today.

The study of **projectile motion** was revolutionized by **Galileo Galilei (1564–1642)**, who showed around 1608 that a projectile follows a *parabolic* path by decomposing it into independent horizontal and vertical motions. **Uniform circular motion** was analysed by **Christiaan Huygens (1629–1695)**, who derived the formula for centripetal acceleration ( $v^2/r$ ) in his 1673 work *Horologium Oscillatorium*.

## 3.1 Scalars and Vectors

### 3.1.1 Definitions

#### Definitions

**Scalar Quantity:** A physical quantity that has *only magnitude* (numerical value with unit) and no direction. Scalars obey ordinary algebra.

**Vector Quantity:** A physical quantity that has *both magnitude and direction*. Vectors obey the laws of vector algebra (triangle law, parallelogram law, etc.).

### 3.1.2 Examples

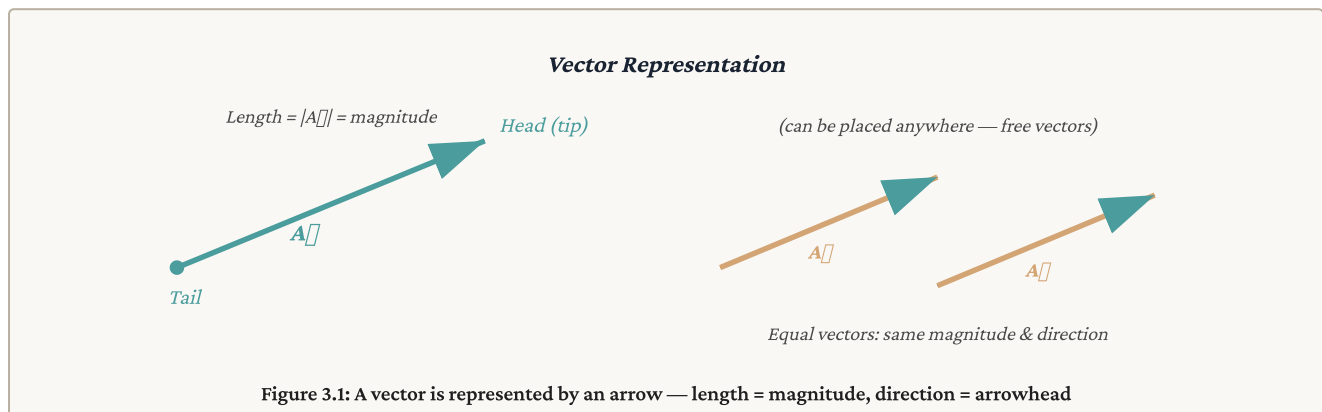
| Scalar Quantities        | Vector Quantities                            |
|--------------------------|----------------------------------------------|
| Mass (5 kg)              | Displacement (5 m east)                      |
| Distance (10 m)          | Velocity (20 m/s north)                      |
| Speed (60 km/h)          | Acceleration (9.8 m/s <sup>2</sup> downward) |
| Time (3 s)               | Force (10 N at 30°)                          |
| Temperature (300 K)      | Momentum (kg m/s)                            |
| Energy / Work (50 J)     | Torque (N m)                                 |
| Density, Pressure, Power | Electric field, Gravitational field          |

### ⚠ Important Distinction

Not every quantity with direction is a vector! To be a true vector, the quantity must also obey the **laws of vector addition** (triangle/parallelogram law). For example, electric current has direction but is a scalar because currents at a junction don't add as vectors — they add algebraically (Kirchhoff's law).

## 3.2 Vector Representation

Vectors are represented by arrows. The **length** of the arrow represents the magnitude (drawn to scale), and the **arrowhead** indicates the direction. In text, a vector is written as  $\vec{A}$  (with an arrow above) or in bold as **A**. Its magnitude is written as  $|\vec{A}|$  or simply  $A$ .



### 3.2.1 Types of Vectors

#### Important Vector Types

- **Equal vectors:** Same magnitude AND same direction
- **Negative vector ( $-\vec{A}$ ):** Same magnitude as  $\vec{A}$  but opposite direction
- **Zero vector ( $\vec{0}$ ):** Magnitude = 0, direction undefined. Example: displacement for a round trip
- **Unit vector ( $\hat{A}$ ):** A vector with magnitude = 1, used to specify direction only.  $\hat{A} = \vec{A} / |\vec{A}|$
- **Co-linear vectors:** Vectors along the same line (parallel or anti-parallel)
- **Co-planar vectors:** Vectors lying in the same plane

### 3.2.2 Unit Vectors

#### Definition

**Unit Vector:** A vector of magnitude 1, used purely to indicate direction. The unit vector in the direction of  $\vec{A}$  is:  $\hat{A} = \vec{A} / |\vec{A}|$ .  
So  $\vec{A} = |\vec{A}| \times \hat{A}$  (magnitude  $\times$  direction).

The standard Cartesian unit vectors are:

#### CARTESIAN UNIT VECTORS

$\hat{i}$  = unit vector along the positive x-axis  
 $\hat{j}$  = unit vector along the positive y-axis  
 $\hat{k}$  = unit vector along the positive z-axis

$$|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$$



## Josiah Willard Gibbs — Father of Modern Vector Analysis

1839–1903, United States

Gibbs, a Yale professor, developed the modern system of vector algebra in the 1880s, introducing the dot product and cross product as we know them. He simplified Hamilton's complex quaternion system into the practical  $\hat{i}, \hat{j}, \hat{k}$  framework used in every physics textbook today. His unpublished lecture notes, circulated among students, became one of the most influential works in mathematical physics. Gibbs also made foundational contributions to thermodynamics and statistical mechanics.

### 3.2.3 Position Vector

#### Definition

**Position Vector ( $\mathbf{r}$ ):** A vector drawn from the origin  $O$  to the location of a point  $P$  in space. In 2D:  $\mathbf{r} = x\hat{i} + y\hat{j}$ . In 3D:  $\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$ . Its magnitude is  $r = |\mathbf{r}| = \sqrt{x^2 + y^2}$  in 2D.

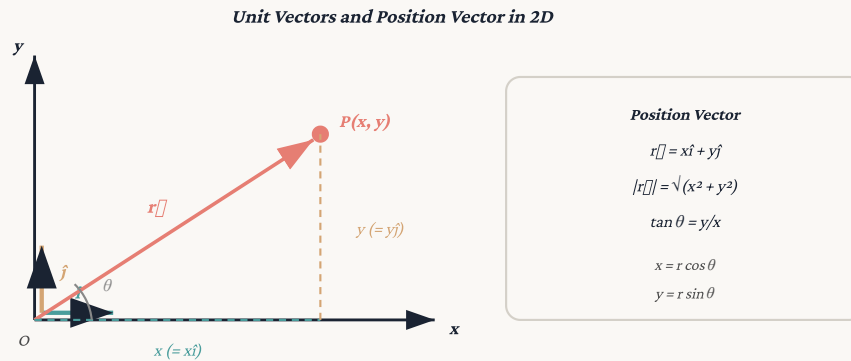


Figure 3.2: Unit vectors  $\hat{i}, \hat{j}$  and the position vector  $\mathbf{r} = x\hat{i} + y\hat{j}$  of point  $P$

## 3.3 Vector Operations

### 3.3.1 Vector Addition — Triangle Law

#### Triangle Law of Vector Addition

If two vectors are represented in magnitude and direction by **two sides of a triangle taken in order**, then the resultant is represented in magnitude and direction by the **third side taken in the opposite order** (closing the triangle).

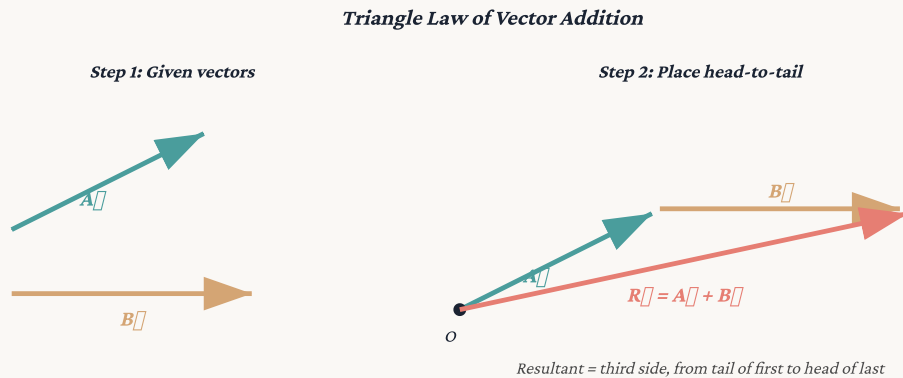


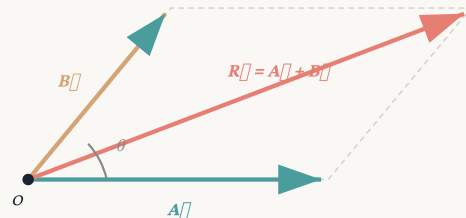
Figure 3.3: Triangle law — place vectors head-to-tail; resultant closes the triangle

### 3.3.2 Vector Addition — Parallelogram Law

#### ▣ Parallelogram Law of Vector Addition

If two vectors are represented in magnitude and direction by **two adjacent sides of a parallelogram** drawn from the same point, then the resultant is represented by the **diagonal** of the parallelogram passing through that point.

Parallelogram Law of Vector Addition



**Magnitude of  $R$ :**

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

**Direction of  $R$ :**

$$\tan \alpha = B \sin \theta / (A + B \cos \theta)$$

( $\alpha$  = angle of  $R$  with  $A$ )

( $\theta$  = angle between  $A$  and  $B$ )

Figure 3.4: Parallelogram law — resultant is the diagonal from the common point

RESULTANT OF TWO VECTORS (PARALLELOGRAM LAW)

$$\text{Magnitude: } R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\text{Direction: } \tan \alpha = B \sin \theta / (A + B \cos \theta)$$

where  $\theta$  = angle between  $A$  and  $B$ ,  $\alpha$  = angle of  $R$  with  $A$

#### Special Cases

- $\theta = 0^\circ$  (same direction):  $R = A + B$  (maximum),  $\alpha = 0^\circ$
- $\theta = 180^\circ$  (opposite):  $R = |A - B|$  (minimum), along the larger vector
- $\theta = 90^\circ$  (perpendicular):  $R = \sqrt{A^2 + B^2}$ ,  $\tan \alpha = B/A$
- $A = B$ :  $R = 2A \cos(\theta/2)$ , and  $R$  bisects the angle between them

#### 💡 Solved Example 3.1

**Q:** Two forces of 6 N and 8 N act at a point. Find the resultant when the angle between them is (a)  $0^\circ$ , (b)  $90^\circ$ , (c)  $180^\circ$ , (d)  $60^\circ$ .

**Solution:**  $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$ ,  $A = 6$ ,  $B = 8$

(a)  $\theta = 0^\circ$ :  $R = \sqrt{36 + 64 + 96} = \sqrt{196} = 14 \text{ N}$

(b)  $\theta = 90^\circ$ :  $R = \sqrt{36 + 64 + 0} = \sqrt{100} = 10 \text{ N}$

(c)  $\theta = 180^\circ$ :  $R = \sqrt{36 + 64 - 96} = \sqrt{4} = 2 \text{ N}$

(d)  $\theta = 60^\circ$ :  $R = \sqrt{36 + 64 + 2 \times 6 \times 8 \times \cos 60^\circ} = \sqrt{100 + 48} = \sqrt{148} \approx 12.2 \text{ N}$

### 3.3.3 Vector Subtraction

VECTOR SUBTRACTION

$$A - B = A + (-B)$$

Reverse the direction of  $B$  to get  $-B$ , then add  $A$  and  $-B$  using triangle/parallelogram law.

$$\text{Magnitude: } |A - B| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

### 3.3.4 Multiplication of a Vector by a Scalar

#### Scalar Multiplication

If  $\lambda$  is a scalar and  $\mathbf{A}$  is a vector, then  $\lambda\mathbf{A}$  is a vector whose:

- Magnitude =  $|\lambda| \times |\mathbf{A}|$
  - Direction = same as  $\mathbf{A}$  if  $\lambda > 0$ ; opposite if  $\lambda < 0$
- Examples:  $3\mathbf{A}$  has 3 times the magnitude in the same direction.  $-2\mathbf{A}$  has 2 times the magnitude in the opposite direction.

### 3.3.5 Cross Product (Vector Product)

#### Definition

**Cross Product (Vector Product):** The cross product of two vectors  $\mathbf{A}$  and  $\mathbf{B}$  is a *vector* quantity defined as:

$$\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{n}$$

where  $\theta$  is the angle between  $\mathbf{A}$  and  $\mathbf{B}$ , and  $\hat{n}$  is the unit vector perpendicular to the plane containing  $\mathbf{A}$  and  $\mathbf{B}$  (direction given by the **right-hand rule**).

CROSS PRODUCT — COMPONENT FORM

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

#### Properties of the Cross Product

- **Anti-commutative:**  $\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$
- **Distributive:**  $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$
- $|\mathbf{A} \times \mathbf{B}| = AB \sin \theta$  = area of parallelogram formed by  $\mathbf{A}$  and  $\mathbf{B}$
- $\hat{i} \times \hat{j} = \hat{k}$ ,  $\hat{j} \times \hat{k} = \hat{i}$ ,  $\hat{k} \times \hat{i} = \hat{j}$  (cyclic order)
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \mathbf{0}$  (parallel vectors)
- If  $\mathbf{A} \times \mathbf{B} = \mathbf{0}$  and neither is zero  $\rightarrow \mathbf{A} \parallel \mathbf{B}$  (parallel)
- If  $\theta = 90^\circ$ :  $|\mathbf{A} \times \mathbf{B}| = AB$  (maximum)
- **Right-hand rule:** Curl fingers from  $\mathbf{A}$  to  $\mathbf{B}$ ; thumb points in direction of  $\mathbf{A} \times \mathbf{B}$

**Physical applications:** The cross product is essential for defining **torque** ( $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ ), **angular momentum** ( $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ ), and the **magnetic force** ( $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ ). It always produces a vector perpendicular to both input vectors.

#### Solved Example 3.10

**Q:** Find  $\mathbf{A} \times \mathbf{B}$  if  $\mathbf{A} = 2\hat{i} + 3\hat{j} + \hat{k}$  and  $\mathbf{B} = \hat{i} - \hat{j} + 2\hat{k}$ .

**Solution:** Using the determinant method:

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (3 \times 2 - 1 \times (-1))\hat{i} - (2 \times 2 - 1 \times 1)\hat{j} + (2 \times (-1) - 3 \times 1)\hat{k} \\ &= (6 + 1)\hat{i} - (4 - 1)\hat{j} + (-2 - 3)\hat{k} = 7\hat{i} - 3\hat{j} - 5\hat{k} \end{aligned}$$

## 3.4 Resolution of Vectors

#### Definition

**Resolution:** The process of splitting a single vector into two (or more) component vectors along chosen directions, usually along perpendicular axes (x and y). This is the reverse of vector addition.

If  $\vec{A}$  makes angle  $\theta$  with the positive x-axis:

$$\text{x-component: } A_x = A \cos \theta$$

$$\text{y-component: } A_y = A \sin \theta$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} = (A \cos \theta) \hat{i} + (A \sin \theta) \hat{j}$$

$$\text{Magnitude: } A = \sqrt{A_x^2 + A_y^2}$$

$$\text{Direction: } \tan \theta = A_y / A_x$$

#### Standard Trigonometric Values (Frequently Used in Problems)

$$\rightarrow \sin 0^\circ = 0, \cos 0^\circ = 1$$

$$\rightarrow \sin 30^\circ = 0.5, \cos 30^\circ = \sqrt{3}/2 \approx 0.866$$

$$\rightarrow \sin 37^\circ \approx 0.6, \cos 37^\circ \approx 0.8 \text{ (3-4-5 triangle)}$$

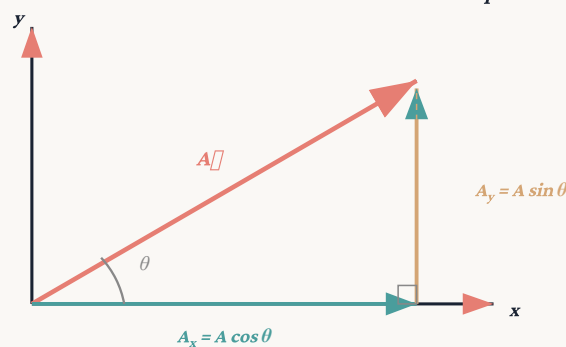
$$\rightarrow \sin 45^\circ = \cos 45^\circ = 1/\sqrt{2} \approx 0.707$$

$$\rightarrow \sin 53^\circ \approx 0.8, \cos 53^\circ \approx 0.6 \text{ (3-4-5 triangle)}$$

$$\rightarrow \sin 60^\circ = \sqrt{3}/2 \approx 0.866, \cos 60^\circ = 0.5$$

$$\rightarrow \sin 90^\circ = 1, \cos 90^\circ = 0$$

#### Resolution of Vector into Components



#### Resolution Formulas

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1}(A_y / A_x)$$

Figure 3.5: Resolving vector  $\vec{A}$  into x and y components

#### 💡 Solved Example 3.2

**Q:** A force of 10 N acts at  $60^\circ$  to the horizontal. Find its horizontal and vertical components.

**Solution:**

$$F_x = F \cos 60^\circ = 10 \times 0.5 = 5 \text{ N}$$

$$F_y = F \sin 60^\circ = 10 \times (\sqrt{3}/2) = 10 \times 0.866 = 8.66 \text{ N}$$

$$\text{Check: } \sqrt{(5^2 + 8.66^2)} = \sqrt{(25 + 75)} = \sqrt{100} = 10 \text{ N } \checkmark$$

## 3.5 Vector Addition — Analytical (Component) Method

The most practical method for adding vectors is to resolve each vector into components, add corresponding components, and then recombine.

If  $\vec{R} = \vec{A} + \vec{B}$ , then:

$$R_x = A_x + B_x = A \cos \alpha + B \cos \beta$$

$$R_y = A_y + B_y = A \sin \alpha + B \sin \beta$$

$$\text{Magnitude: } R = \sqrt{R_x^2 + R_y^2}$$

$$\text{Direction: } \theta = \tan^{-1}(R_y / R_x)$$

#### Solved Example 3.3

**Q: Find the resultant of  $\vec{A} = 3\hat{i} + 4\hat{j}$  and  $\vec{B} = -2\hat{i} + 5\hat{j}$ .**

**Solution:**

$$\vec{R} = \vec{A} + \vec{B} = (3 - 2)\hat{i} + (4 + 5)\hat{j} = \hat{i} + 9\hat{j}$$

$$\text{Magnitude: } R = \sqrt{(1^2 + 9^2)} = \sqrt{(1 + 81)} = \sqrt{82} \approx \mathbf{9.06 \text{ units}}$$

$$\text{Direction: } \theta = \tan^{-1}(9/1) = \tan^{-1}(9) \approx \mathbf{83.7^\circ} \text{ with x-axis}$$

#### Solved Example 3.4

**Q: Two vectors of magnitudes 5 and 12 are perpendicular. Find their resultant.**

$$\text{Solution: } \theta = 90^\circ \rightarrow R = \sqrt{(A^2 + B^2)} = \sqrt{(25 + 144)} = \sqrt{169} = \mathbf{13 \text{ units}}$$

$$\text{Direction: } \alpha = \tan^{-1}(12/5) = \tan^{-1}(2.4) \approx \mathbf{67.4^\circ} \text{ with the 5-unit vector}$$

## 3.6 Projectile Motion

### Definition

**Projectile:** Any object thrown into the air with some initial velocity and then allowed to move under the influence of gravity alone (air resistance neglected). The path followed by a projectile is called its **trajectory**.

### Galileo's Revolutionary Discovery of Projectile Motion

**Before Galileo:** Medieval scholars believed a cannonball travels in a straight line until its "impetus" runs out, then falls vertically. This was wrong but widely accepted for centuries.

**Galileo's Breakthrough (c. 1608):** Through careful experiments rolling balls off tables and tracking where they landed, Galileo discovered two crucial facts: (1) the horizontal and vertical motions are *completely independent* of each other, and (2) the resulting path is a **parabola**. He published this in his masterwork *Discourses and Mathematical Demonstrations Relating to Two New Sciences* (1638). This was one of the first applications of the **superposition principle** — analyzing a complex motion by breaking it into simpler, independent components — an idea that remains central to physics today.

The key insight is that projectile motion can be analyzed as **two independent, simultaneous motions**:

### Decomposition of Projectile Motion

→ **Horizontal (x):** Uniform velocity ( $a_x = 0$ ) →  $x = u_x t = (u \cos \theta) t$

→ **Vertical (y):** Uniformly accelerated ( $a_y = -g$ ) →  $y = u_y t - \frac{1}{2}gt^2 = (u \sin \theta) t - \frac{1}{2}gt^2$

→ These two motions are completely **independent** of each other

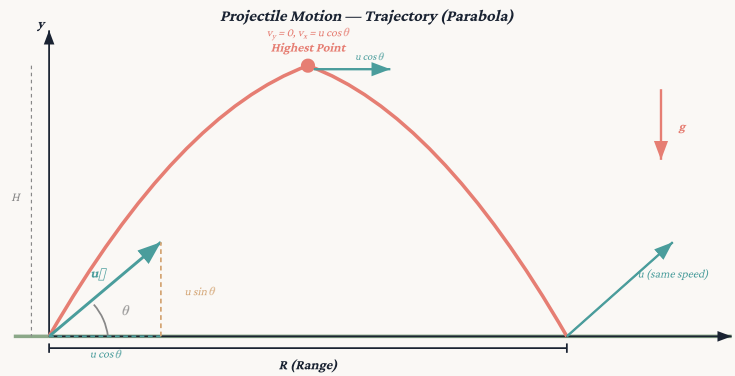


Figure 3.6: Projectile trajectory — a parabola with horizontal range R and maximum height H

### 3.6.1 Projectile Motion Equations

For a projectile launched with speed  $u$  at angle  $\theta$  from the horizontal ground:

TRAJECTORY (EQUATION OF PATH)

$$y = x \tan \theta - gx^2 / (2u^2 \cos^2 \theta) \quad (\text{a parabola})$$

TIME OF FLIGHT

$$T = 2u \sin \theta / g$$

MAXIMUM HEIGHT

$$H = u^2 \sin^2 \theta / (2g)$$

HORIZONTAL RANGE

$$R = u^2 \sin 2\theta / g$$

#### Important Results

- **Maximum range** occurs at  $\theta = 45^\circ$ :  $R_{\max} = u^2/g$
- **Complementary angles** give the same range:  $R(\theta) = R(90^\circ - \theta)$ . E.g.,  $30^\circ$  and  $60^\circ$  give the same range
- At the highest point:  $v_y = 0$ ,  $v_x = u \cos \theta$  (horizontal velocity never changes)
- Time to reach max height =  $T/2 = u \sin \theta / g$
- Speed at any time:  $v = \sqrt{v_x^2 + v_y^2}$  where  $v_x = u \cos \theta$ ,  $v_y = u \sin \theta - gt$

#### Solved Example 3.5

**Q:** A ball is thrown at 20 m/s at  $30^\circ$  to the horizontal. Find (a) time of flight, (b) max height, (c) range. ( $g = 10 \text{ m/s}^2$ )

**Solution:**  $u = 20 \text{ m/s}$ ,  $\theta = 30^\circ$ ,  $\sin 30^\circ = 0.5$ ,  $\cos 30^\circ = \sqrt{3}/2$ ,  $\sin 60^\circ = \sqrt{3}/2$

(a)  $T = 2u \sin \theta / g = 2(20)(0.5)/10 = 2 \text{ s}$

(b)  $H = u^2 \sin^2 \theta / 2g = 400 \times 0.25 / 20 = 5 \text{ m}$

(c)  $R = u^2 \sin 2\theta / g = 400 \times \sin 60^\circ / 10 = 400 \times 0.866/10 = 34.64 \text{ m}$

#### Solved Example 3.6

**Q:** Show that the range is same for  $\theta = 30^\circ$  and  $\theta = 60^\circ$  for the same launch speed.

**Solution:**

$$R = u^2 \sin 2\theta / g$$

For  $\theta = 30^\circ$ :  $R_1 = u^2 \sin 60^\circ / g = u^2(\sqrt{3}/2) / g$

For  $\theta = 60^\circ$ :  $R_2 = u^2 \sin 120^\circ / g = u^2 \sin(180^\circ - 60^\circ) / g = u^2(\sqrt{3}/2) / g$

$$\mathbf{R_1 = R_2} \checkmark \text{ (since } \sin 60^\circ = \sin 120^\circ \text{)}$$

In general,  $\sin 2\theta = \sin 2(90^\circ - \theta) = \sin(180^\circ - 2\theta) = \sin 2\theta$ . So  $R(\theta) = R(90^\circ - \theta)$ .

### Solved Example 3.7

**Q:** A projectile has a range of 500 m when launched at  $45^\circ$ . What is the launch speed? ( $g = 10 \text{ m/s}^2$ )

**Solution:** At  $\theta = 45^\circ$ ,  $R = u^2/g$  (since  $\sin 90^\circ = 1$ )

$$500 = u^2/10 \rightarrow u^2 = 5000 \rightarrow u = \sqrt{5000} \approx 70.7 \text{ m/s}$$

## 3.7 Uniform Circular Motion

### Definitions

**Circular Motion:** Motion of an object along a circular path.

**Uniform Circular Motion (UCM):** Motion along a circular path with *constant speed*. The *direction* of velocity changes continuously, so the object is accelerating even though its speed is constant.



### Christiaan Huygens — Pioneer of Circular Motion

1629–1695, Netherlands

Huygens derived the formula for centripetal acceleration ( $a = v^2/r$ ) in his 1673 work *Horologium Oscillatorium*, originally motivated by his invention of the pendulum clock — he needed to understand circular motion to design better timekeeping devices. Newton later used Huygens' centripetal acceleration result as a key ingredient in deriving the law of gravitation, combining  $v^2/r$  with Kepler's Third Law to show that gravity follows an inverse-square law. Huygens also proposed the wave theory of light and discovered Saturn's rings.

### 3.7.1 Key Quantities in Circular Motion

#### CIRCULAR MOTION QUANTITIES

Angular displacement:  $\theta$  (in radians)

Angular velocity:  $\omega = \theta/t = 2\pi/T = 2\pi f$  (rad/s)

Time period:  $T = 2\pi/\omega = 1/f$  (seconds)

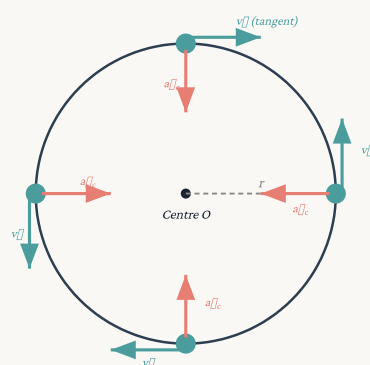
Frequency:  $f = 1/T = \omega/2\pi$  (Hz)

Speed:  $v = \omega r = 2\pi r/T$

### 3.7.2 Centripetal Acceleration

In UCM, the speed is constant but the *velocity direction changes continuously*. This change in velocity means there is an acceleration — directed toward the centre of the circle. This is called **centripetal acceleration** ("centre-seeking").

#### Uniform Circular Motion — Centripetal Acceleration



#### Key Facts (UCM)

- $v(t)$  is always tangent to the circle
- $a(t)_c$  always points toward the centre
- $v(t) \perp a(t)_c$  (always)
- Speed  $|v| = \text{constant}$
- $|a_c| = \text{constant}$

Figure 3.7: In UCM, velocity is tangential and centripetal acceleration points toward the centre

## CENTRIPETAL ACCELERATION

$$a_c = v^2/r = \omega^2 r = v\omega$$

Direction: Always toward the centre of the circle

$$\text{Centripetal Force: } F_c = ma_c = mv^2/r$$

 **Solved Example 3.8**

**Q:** A car moves at 36 km/h around a circular track of radius 100 m. Find the centripetal acceleration.

**Solution:**  $v = 36 \times 5/18 = 10 \text{ m/s}$ ,  $r = 100 \text{ m}$

$a_c = v^2/r = 100/100 = 1 \text{ m/s}^2$  (directed toward the centre)

 **Solved Example 3.9**

**Q:** A stone is tied to a string of length 0.5 m and whirled in a horizontal circle at 4 rev/s. Find (a) angular velocity, (b) linear speed, (c) centripetal acceleration.

**Solution:**

(a)  $\omega = 2\pi f = 2\pi \times 4 = 8\pi \text{ rad/s} \approx 25.13 \text{ rad/s}$

(b)  $v = \omega r = 8\pi \times 0.5 = 4\pi \approx 12.57 \text{ m/s}$

(c)  $a_c = \omega^2 r = (8\pi)^2 \times 0.5 = 64\pi^2 \times 0.5 = 32\pi^2 \approx 315.8 \text{ m/s}^2$

 **Common Misconceptions about Circular Motion**

- "An object in UCM is not accelerating" — **Wrong!** Velocity direction changes, so there IS acceleration (centripetal)
- "Centripetal force is a new type of force" — **Wrong!** It is the NET force directed toward the centre, provided by tension, gravity, friction, normal force, etc.
- "The object flies outward" — No! If the centripetal force is removed, the object flies off **tangentially** (not radially outward)

 **Chapter Summary — Must Remember**

- Scalars have magnitude only; vectors have magnitude AND direction
- Unit vector:  $\hat{A} = A/|A|$ , magnitude = 1. Standard:  $\hat{i}, \hat{j}, \hat{k}$
- Position vector:  $\vec{r} = x\hat{i} + y\hat{j}$ ;  $|\vec{r}| = \sqrt{x^2 + y^2}$
- Triangle law: place vectors head-to-tail; resultant = tail of first to head of last
- Parallelogram law:  $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$
- Special cases: same direction  $R = A+B$ ; opposite  $R = |A-B|$ ; perpendicular  $R = \sqrt{A^2+B^2}$
- Resolution:  $A_x = A \cos \theta$ ,  $A_y = A \sin \theta$
- Projectile = horizontal uniform motion + vertical free fall
- $T = 2u \sin \theta / g$ ,  $H = u^2 \sin^2 \theta / 2g$ ,  $R = u^2 \sin 2\theta / g$
- Max range at  $45^\circ$ ; complementary angles ( $\theta$  and  $90^\circ - \theta$ ) give same range
- UCM: speed constant, velocity direction changes continuously
- Centripetal acceleration:  $a_c = v^2/r = \omega^2 r$ , always toward centre

 **Common Mistakes to Avoid**

- Adding vectors as ordinary numbers — must use vector laws
- Confusing resolution (splitting) with addition (combining)
- Using sin where cos is needed in component resolution
- Forgetting that horizontal velocity is constant in projectile motion
- Thinking velocity is zero at the highest point of a projectile — only  $v_y = 0$ ;  $v_x \neq 0$
- Confusing centripetal (toward centre) with centrifugal (fictitious, outward)
- Treating centripetal force as a separate force rather than the net inward force

## Practice Questions (Unsolved)

### Conceptual Questions

1. Can two vectors of different magnitudes give zero resultant? What about three?
2. Is velocity at highest point of a projectile zero? Explain.
3. Why does object in UCM have acceleration despite constant speed?
4. Is centripetal force a new type of force? What provides it in different situations?
5. Why is the cross product of two parallel vectors zero? What does this mean physically?
6. What happens to projectile range if angle changes from  $30^\circ$  to  $60^\circ$ ?

### Short Answer Questions (2-3 Marks)

1. State parallelogram law. Derive the resultant formula.
2. Show that complementary angles give same range in projectile motion.
3. Resolve a force of 20 N at  $37^\circ$  into horizontal and vertical components.
4. Derive the relation between linear and angular velocity.

### Numerical Problems

1. Find resultant of 6 N and 8 N at (a)  $0^\circ$ , (b)  $90^\circ$ , (c)  $60^\circ$ , (d)  $180^\circ$ .
2. Ball at 20 m/s at  $30^\circ$ . Find T, H, and R. ( $g = 10 \text{ m/s}^2$ )
3. Car at 36 km/h on circular track  $r = 100 \text{ m}$ . Find centripetal acceleration.
4. Find resultant of  $A = 3\mathbf{i} + 4\mathbf{j}$  and  $B = -2\mathbf{i} + 5\mathbf{j}$ . Find magnitude and direction.
5. Stone whirled,  $r = 0.5 \text{ m}$  at 4 rev/s. Find  $\omega$ ,  $v$ , and centripetal acceleration.
6. Projectile has range 500 m at  $45^\circ$ . Find launch speed. ( $g = 10 \text{ m/s}^2$ )

### Assertion-Reason Questions

(a) Both true, R explains A (b) Both true, R not explanation (c) A true, R false (d) A false, R true

1. **A:** Range is same for  $30^\circ$  and  $60^\circ$ . **R:**  $R = u^2 \sin 2\theta / g$  and  $\sin 60^\circ = \sin 120^\circ$ .
2. **A:** In UCM, speed is constant but velocity is not. **R:** Velocity direction changes continuously.
3. **A:** Horizontal velocity is constant in projectile motion. **R:** No horizontal acceleration (air resistance neglected).